Polynomial and Interaction Models

Applied Linear Models I • Lecture 9 of 16

Applied Linear Models I

Today

Enriching the mean function while keeping OLS (still linear in parameters):

- polynomials and **orthogonal** bases;
- **interactions** – one variable's slope depends on another;
- **centering** to tame collinearity;
- the **marginality** principle.

Polynomials: raw powers are collinear

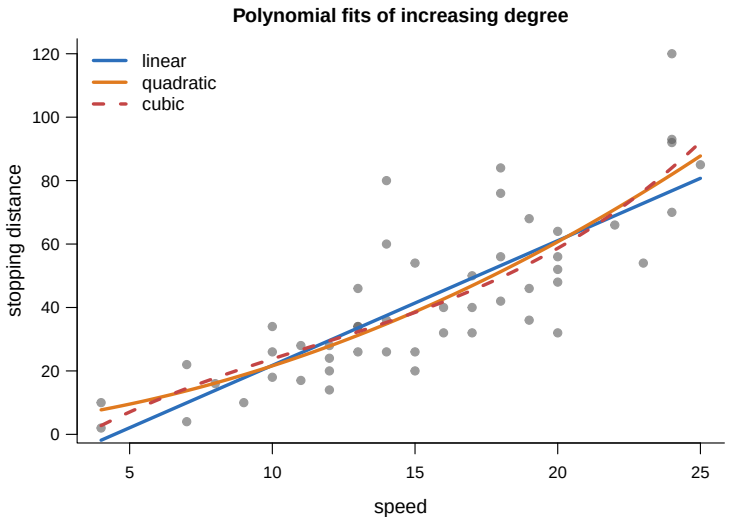
$$y = \beta_0 + \beta_1 x + \cdots + \beta_d x^d + \varepsilon$$

Raw x, x^2, x^3 on `cars`: pairwise correlations > 0.97. `poly(x,d)` returns an **orthogonal** basis (correlations ≈ 0).

Same fit, better conditioning

Raw and orthogonal parameterizations give *identical* fitted values and R^2 (agree to 10^{-8}). Only stability and coefficient meaning differ.

Polynomial fits



Interactions

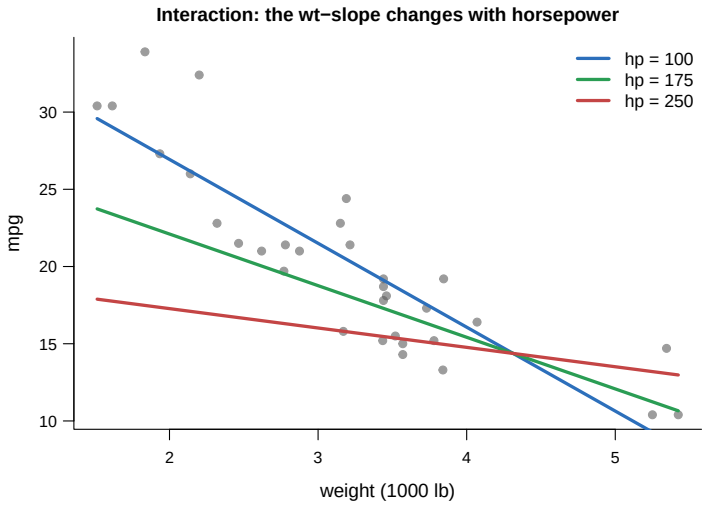
`mpg ~ wt * hp` $=$ `wt + hp + wt:hp`:

$$\frac{\partial\,\mathbb{E}[\text{mpg}]}{\partial\,\text{wt}} = \beta_{\text{wt}} + \beta_{\text{wt:hp}} \cdot \text{hp}.$$

hp	$\partial(\text{mpg})/\partial(\text{wt})$
100	-5.43
175	-3.34
250	-1.26

The weight penalty **weakens** as horsepower rises.

Read interactions from fitted-value plots



The "main effect" β_{wt} is the slope at $hp = 0$ — an extrapolation, unless hp is centered. Interpret

Centering reduces collinearity

Products/powers are mechanically correlated with their parts. Center first:

Term	VIF uncentered	VIF centered
wt	10.3	1.8
hp	19.2	1.8
wt:hp	43.1	1.0

VIFs collapse to ≈ 1; the fit is identical (to 10^{-14}). The cheapest collinearity remedy – make it routine.

Marginality (hierarchy) principle

Rule

If a higher-order term is in the model, keep all lower-order terms it contains: include `wt`, `hp` with `wt:hp`; include x with x^2.

- Interaction-only $\Rightarrow$ effect of x_1 is zero at $x_2 = 0$ – an **origin-dependent** constraint.
- Dropping `hp`'s main effect: adjusted R^2 falls $0.872 \rightarrow 0.773$.

Summary

- `poly()` for stable polynomials; same fit as raw powers.
- Interactions: slope $= \beta_1 + \beta_3 x_2$; use fitted-value plots.
- Center before interacting/powering – VIFs 43 $\rightarrow$ 1, fit unchanged.
- Keep lower-order terms (marginality).

Next

Lecture 10: multicollinearity – detection and remedies (ridge, PCR).